

Nombre de la práctica	set, joins y subconsultas			No.	6
Asignatura:	Taller de Base de Datos	Carrera:	ISIC	Duración de la práctica (Hrs)	4

I. Competencia(s) específica(s): Realizar consultas con set, joins y subconsultas

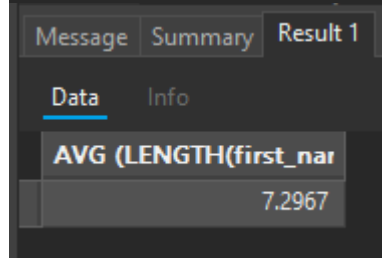
II. Lugar de realización de la práctica (laboratorio, taller, aula u otro): Aula

III. Material empleado:

Computadora

IV. Desarrollo de la práctica:

```
-- Calcular el promedio de la longitud de
-- first_name en la tabla actor
SELECT AVG (LENGTH(first_name)) FROM actor;
```



The screenshot shows a SQL query result in a table with two columns: 'AVG (LENGTH(first_name))' and '7.2967'. The table is titled 'Result 1' and has tabs for 'Message', 'Summary', and 'Data'.

```
-- Seleccionar actores con longitud de
-- first_name mayor al promedio
SELECT * FROM actor WHERE LENGTH(first_name)
> (SELECT AVG(LENGTH(first_name)) FROM actor);
```



Message

Summary

Result 1

Data

Info

actor_id	first_name	last_name	last_update
1	PENELOPE	GUINESS	2006-02-15 04:34:33
4	JENNIFER	DAVIS	2006-02-15 04:34:33
10	CHRISTIAN	GABLE	2006-02-15 04:34:33
54	PENELOPE	PINKETT	2006-02-15 04:34:33
58	CHRISTIAN	AKROYD	2006-02-15 04:34:33
61	CHRISTIAN	NEESON	2006-02-15 04:34:33
70	MICHELLE	MCCONAUGH	2006-02-15 04:34:33

```
-- Encuentra los actores que han participado
-- en películas de la categoría comedy

-- Separado
SELECT category_id FROM category WHERE name = "Comedy";
SELECT film_id FROM film_category WHERE category_id = 5;
```

Message	Summary	Result 1
Data	Info	
category_id		
5		



Message	Summary	Result 1
Data	Info	
film_id		
7		
28		
99		
119		
127		
159		
178		

```
-- Para listas hacemos uso del in y del not in
SELECT actor_id FROM film_actor WHERE film_id IN (7,28,99);
```



```
-- Encuentra los actores que han participado en las peliculas de
la categoria comedy consultas juntas
SELECT actor_id FROM film_actor WHERE film_id IN (SELECT film_id
FROM film_category WHERE category_id = (SELECT category_id FROM
category WHERE name = "Comedy"));
```

```
SELECT first_name, last_name FROM actor WHERE actor_id IN (SELECT actor_id FROM film_actor WHERE film_id IN (SELECT film_id FROM film_category WHERE category_id = (SELECT category_id FROM category WHERE name = "Comedy")));
```



Message

Summary

Result 1

Data

Info

first_name	last_name
JIM	MOSTEL
RICHARD	PENN
OPRAH	KILMER
MENA	HOPPER
MICHAEL	BOLGER
MILLA	KEITEL
FRANCES	DAY-LEWIS

```
-- Encuentra los títulos de las películas que nunca han sido  
alquiladas  
-- Ver id de la tabla rental  
SELECT rental_id FROM rental WHERE rental_id ;
```

MessageSummaryR

DataInfo

rental_id
1
2
3
4
5
6
7

```
-- Inventario  
SELECT film_id FROM inventory WHERE inventory_id IN ( SELECT  
inventory_id FROM rental WHERE rental_id );
```



Message	Summary	Result 1
Data	Info	
film_id		
1		
1		
1		
1		
1		
1		
1		
1		

```
-- Pelicula
SELECT title FROM film WHERE film_id NOT IN (SELECT film_id FROM
inventory WHERE inventory_id IN ( SELECT inventory_id FROM rental
WHERE rental_id ));
```

Message	Summary	Result 1
Data	Info	
title		
ALICE FANTASIA		
APOLLO TEEN		
ARGONAUTS TOWN		
ARK RIDGEMONT		
ARSENIC INDEPENDENCE		
BOONDOCK BALLROOM		
BUTCH PANTHER		



```
-- INNER JOIN cuando las tablas coinciden
SELECT f.title, c.name AS category_name
FROM film AS f
INNER JOIN film_category AS fc
ON f.film_id = fc.film_id
INNER JOIN category AS c
ON fc.category_id = c.category_id
WHERE c.name = "Comedy";
```

Message Summary Result 1

Data Info

title	category_name
AIRPLANE SIERRA	Comedy
ANTHEM LUKE	Comedy
BRINGING HYSTERICAL	Comedy
CAPER MOTIONS	Comedy
CAT CONEHEADS	Comedy
CLOSER BANG	Comedy
CONNECTION MICROCOS	Comedy

```
-- INNER JOIN
SELECT film.title, c.name FROM film
INNER JOIN film_category ON film.film_id = film_category.film_id
JOIN category AS c ON film_category.category_id = c.category_id;
```



Message	Summary	Result 1
Data	Info	
title	name	
AMADEUS HOLY	Action	
AMERICAN CIRCUS	Action	
ANTITRUST TOMATOES	Action	
ARK RIDGEMONT	Action	
BAREFOOT MANCHURIAN	Action	
BERETS AGENT	Action	
BRIDE INTRIGUE	Action	

```
-- LEFT JOIN Toma los elementos de la izquierda
-- Encuentra el titlulo de las peliculas y la cantidad de veces
que han sido alquiladas
SELECT f.title, COUNT(r.rental_id) AS cantidad_rentas
FROM film AS f
LEFT JOIN inventory AS i ON f.film_id = i.film_id
LEFT JOIN rental AS r ON i.inventory_id= r.inventory_id
GROUP BY f.title;
```

Message	Summary	Result 1
Data	Info	
title	cantidad_rentas	
ACADEMY DINOSAUR	23	
ACE GOLDFINGER	7	
ADAPTATION HOLES	12	
AFFAIR PREJUDICE	23	
AFRICAN EGG	12	
AGENT TRUMAN	21	
AIRPLANE SIERRA	15	



```
-- Muestra los nombres de los actores y los títulos de las  
películas en las que han participado
```

```
-- sin CONCAT
```

```
SELECT a.first_name, a.last_name, f.title  
FROM actor AS a  
LEFT JOIN film_actor AS fa ON a.actor_id = fa.actor_id  
LEFT JOIN film AS f ON fa.film_id = f.film_id;
```

Message Summary Result 1

Data Info

first_name	last_name	title
PENELOPE	GUINESS	ACADEMY DINOSAUR
PENELOPE	GUINESS	ANACONDA CONFESSION
PENELOPE	GUINESS	ANGELS LIFE
PENELOPE	GUINESS	BULWORTH COMMANDM
PENELOPE	GUINESS	CHEAPER CLYDE
PENELOPE	GUINESS	COLOR PHILADELPHIA
PENELOPE	GUINESS	ELEPHANT TROJAN

```
-- con CONCAT
```

```
SELECT CONCAT(a.first_name," ",a.last_name) AS full_name, f.title  
FROM actor AS a  
JOIN film_actor AS fa ON a.actor_id = fa.actor_id  
JOIN film AS f ON fa.film_id = f.film_id;
```



Message		Summary		Result 1	
Data		Info			
full_name		title			
PENELOPE GUINESS		ACADEMY DINOSAUR			
CHRISTIAN GABLE		ACADEMY DINOSAUR			
LUCILLE TRACY		ACADEMY DINOSAUR			
SANDRA PECK		ACADEMY DINOSAUR			
JOHNNY CAGE		ACADEMY DINOSAUR			
MENA TEMPLE		ACADEMY DINOSAUR			
WARREN NOLTE		ACADEMY DINOSAUR			

```
-- Operaciones SET
SELECT first_name FROM actor
UNION
SELECT first_name FROM customer;
```

Message		Summary		Result 1	
Data		Info			
first_name					
PENELOPE					
NICK					
ED					
JENNIFER					
JOHNNY					
BETTE					
GRACE					

```
-- Encuentra las películas que no han sido alquiladas
SELECT title FROM film
EXCEPT
SELECT f.title FROM film AS f
JOIN inventory AS i ON f.film_id = i.film_id
JOIN rental AS r ON i.inventory_id = r.inventory_id;
```



```
-- Devuelve las ciudades donde viven los clientes o empleados
sin duplicados
SELECT DISTINCT city FROM city
JOIN address ON city.city_id = address.city_id
JOIN staff ON address.address_id = staff.address_id
```

```
SELECT city FROM city WHERE city_id IN(SELECT city_id FROM address
WHERE address_id IN (SELECT address_id FROM customer));
```



Message	Summary	Result 1
Data	Info	
city		
A Coruña (La C		
Abha		
Abu Dhabi		
Acuña		
Adana		
Addis Abeba		
Aden		

V. Conclusiones:

Las consultas con SET, JOINS y subconsultas permiten manejar y relacionar datos en una base de datos de manera muy potente. Con SET podemos combinar resultados como con UNION, lo que ayuda a integrar información de diferentes fuentes. Los JOINS conectan datos de distintas tablas, creando una vista completa sin duplicar datos, mientras que las subconsultas permiten realizar consultas dentro de otras para filtrar o calcular información específica. En conjunto, estas herramientas hacen que las consultas sean mucho más flexibles y nos permiten obtener datos complejos de manera eficiente.